

Incident Handler's Journal

#1 – Documenting a Cybersecurity Incident

Date: May 19, 2026

Overview: This activity focused on documenting a ransomware incident affecting a healthcare company. The incident followed two major phases: Detection and Analysis, followed by Containment, Eradication, and Recovery. The organization identified the attack, sought technical assistance, and temporarily shut down systems to minimize further damage.

Tools Used: No specific tools were used during this activity.

The 5 W's:

- Who: An organized group of cybercriminals.
- What: A ransomware attack targeting company systems.
- Where: A healthcare organization.
- When: Tuesday at 9:00 a.m.
- Why: Attackers gained access through a phishing email, encrypted critical files, and demanded ransom payment.

Additional Notes: This activity highlighted the importance of phishing awareness, incident response planning, and timely communication during cybersecurity incidents.

#2 – Analyzing a Packet Capture File

Date: 25 July 2024

Overview: This exercise introduced packet analysis using Wireshark. I explored captured network traffic to better understand how analysts investigate suspicious communication and identify abnormal behavior.

Tools Used: Wireshark – a network protocol analyzer used to capture and inspect network traffic through a graphical user interface.

The 5 W's:

- Who: N/A
- What: Packet capture analysis activity.
- Where: Simulated lab environment.
- When: During coursework activity.
- Why: To learn how network traffic analysis supports threat detection.

Additional Notes: Although the interface initially felt overwhelming, the activity helped me understand the value of packet analysis in cybersecurity investigations.

#3 – Capturing My First Packet

Date: 25 July 2024

Overview: This activity focused on capturing live network traffic using tcpdump. It provided practical experience with command-line tools and network packet inspection.

Tools Used: tcpdump – a command-line network traffic analyzer used for capturing, filtering, and analyzing packets.

The 5 W's:

- Who: N/A
- What: Network packet capture exercise.
- Where: Simulated lab environment.
- When: During coursework activity.
- Why: To develop foundational skills in network traffic monitoring and analysis.

Additional Notes: Learning command-line syntax was challenging at first, but repeating the steps and reviewing the instructions carefully improved my confidence.

#4 – Investigating a Suspicious File Hash

Date: 27 July 2024

Overview: In this scenario, I investigated a suspicious SHA-256 file hash that triggered an alert within a financial services organization's SOC. The exercise focused on validating whether the detected file represented a genuine threat.

Tools Used: VirusTotal – an investigative platform used to analyze suspicious files, URLs, and indicators of compromise.

The 5 W's:

- Who: An unidentified malicious actor.
- What: A malicious email attachment containing a suspicious file hash.
- Where: An employee workstation at a financial services company.
- When: Alert generated at 1:20 p.m.
- Why: The employee downloaded and executed a malicious attachment received through email.

Additional Notes: This activity reinforced the importance of security awareness training, email filtering, and proactive threat investigation.

Reflections & Key Takeaways

Most Challenging Activity: Using tcpdump was the most challenging part of the course because I was unfamiliar with command-line syntax. However, repeating the exercises and carefully reviewing the instructions helped me improve significantly.

Growth in Incident Response Knowledge: This course strengthened my understanding of incident detection and response by introducing incident lifecycles, response planning, investigative processes, and cybersecurity tools used by analysts.

Most Enjoyable Concept: I especially enjoyed learning about network traffic analysis. Capturing and analyzing traffic in real time was both challenging and rewarding, and it increased my interest in cybersecurity investigations.